MILITARY SPECIFICATION SHEET

ELECTRON TUBE, POWER
1/ TYPE 7034

(L)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Tetrode

See figure 1

Mounting position: Any

Weight: 4 ounces nominal

ABSOLUTE RATINGS: F1 = 150 MHz; F2 = 500 MHz

Parameter:	Ef	Eb	Ec1	Ec2	Ehk	Ib	Pg1	Pg2	Pp
Unit:	Vac (see note 2)	Vdc	Vdc	Vdc	Vdc	mAdc	W	W	W
Maximum:									
C Tig (Up to 150 MHz)	---	2.000	-250	300	±150	250	2	12	250
C Tig (150 to 500 MHz)	---	1.250	-250	300	±150	250	2	12	250
TEST CONDITIONS:	6.0	2.000	Adj	300	0	150	---	---	---

ABSOLUTE RATINGS F1 = 150 MHz; F2 = 500 MHz

Parameter:	T(base seal)	T(anode seal)	T(anode core)	tk
Unit:	C (see note 1)	C (see note 1)	C (See note 1)	sec (min)
Maximum:				
C Tig (Up to 150 MHz)	175	200	250	30
C Tig (150 to 500 MHz)	175	200	250	30
TEST CONDITIONS:	See note 3	---	---	120

GENERAL:

Qualification - Required

1/ See note 11

(L) denotes changes

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1								
1231	Pulsing emission	7	eb = ec1 = ec2 = 850 v	0.65	II	is	30	---	a
1256	Electrode current (screen)	-	Eb = 1,000 Vdc	0.65	II	Ic2	-5	+3	mAdc
1261	Electrode voltage (grid)	-	Eb = 1,000 Vdc	0.65	II	Ec1	-32	-45	Vdc
1266	Total grid current	8	Ib = 125 mAdc	0.65	II	Ic1	---	-15	µAdc
1372	Current division (method A, long pulse)	-	Eb = Ec2 = 250 Vdc; Ec1 = -100 Vdc; egk ib = 1.0 a; prr = 11.0 ±1.0; tp = 4,500 µs (min)	0.65	II	egk ic1 ic2	8 --- ---	18 250 260	v ma ma
1266	Primary grid emission (control)	-	Ic1 = 70 mAdc; t = 15; anode and screen grid floating	0.65	II	Iag1	---	-25	µAdc
1266	Primary grid emission (screen)	-	Ec1 = 0; Ic2 = 100 mAdc; t = 15; anode floating	0.65	II	Iag2	---	-250	µAdc
1301	Heater current	-		0.65	II	If	2.30	2.90	Aac
	Quality conformance inspection, part 2								
1031	Low-frequency vibration	-	No voltages applied	---	---	---	---	---	---
(L) 1042	Shock, specified pulse	-	No voltages applied; accel = 15 G peak (min); D = 11 ±2 ms half-sine wave	---	---	---	---	---	---
---	Shock, and low-frequency vibration end points:								
1261	Electrode voltage (grid)	-		---	---	Ec1	-32	-45	Vdc
1266	Total grid current	-		---	---	Ic1	---	-15	µAdc
(L) 2214	RF useful power output (1)	-	F = 150 MHz (min); Ec1 = -90 Vdc; Ic1 = 25 mAdc (max); Eg1/Ib = 250 mAdc	---	---	Po	225	---	W (useful)
(L) 2214	RF useful power output (2)	9	Class C amplifier; Ef = 5.5 Vac; F = 460 to 490 MHz; Eb = 1,250 Vdc; Ec1 = -90 Vdc, Ec2 = 250 to 300 Vdc; Ic1 = 20 mAdc (max); Eg1/Ib = 250 mAdc	---	---	Po	145	---	W (useful)

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 2 -Continued								
1331	Direct-interelectrode capacitance	-	EIA standard shield No. 320 and No. 321, or equal	---	---	C_{gk} C_{in} C_{out}	--- 14.5 4.0	0.05 17.0 4.8	pF pF pF
1336	Heater-cathode leakage	-	Ehk = +150 Vdc Ehk = -150 Vdc	---	---	Ihk Ihk	--- ---	150 150	µAdc µAdc
	Quality conformance inspection, part 3								
(L) ---	Life test (1)	-	Group C: rf useful power output (1); t = 500 hours	---	---	---	---	---	---
---	Life-test (1) end points:								
1231	Pulsing emission	-		---	---	Is	21	---	a
1266	Primary grid emission (control)	-		---	---	Isg1	---	-100	µAdc
1266	Primary grid emission (screen)	-		---	---	Isg2	---	-250	µAdc
1336	Heater-cathode leakage	-	Ehk = +150 Vdc Ehk = -150 Vdc	---	---	Ihk Ihk	--- ---	150 150	µAdc µAdc
---	Life test (2)	-	Group C: Ec1 = Ec2 = Eb = 0; t = 500 hours; Ef = 6.6 Vac	---	---	---	---	---	---
---	Life-test (2) end points:								
1366	Interelement leakage resistance, cold	.10	Rs = 2.5 Meg E = 100 Vdc: g1 negative	---	---	Rg1k	10	---	Meg
			E = 500 Vdc: g2 positive	---	---	Rg1g2	10	---	Meg
			E = 500 Vdc: g2 positive	---	---	Rg2k	10	---	Meg
	Periodic-check tests								
(L) ---	Life test (3)	-	RF useful power output (2); t = 500 hours	---	---	---	---	---	---
---	Life-test (3) end points:								
1231	Pulsing emission	-		---	---	Is	21	---	a
1266	Primary grid emission (control)	-		---	---	Isg1	---	-100	µAdc
1266	Primary grid emission (screen)	-		---	---	Isg2	---	-250	µAdc
1336	Heater-cathode leakage	-	Ehk = +150 Vdc Ehk = -150 Vdc	---	---	Ihk Ihk	--- ---	150 150	µAdc µAdc
1155	Coolant pressure drop versus coolant flow (forced air)	4	No voltages applied	---	---	---	---	0.60	inches of water

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMIT MIN	LIMIT MAX	UNIT
	Periodic-check tests -Continued								
1143	Forced cooling	5	Eb = 1,000 Vdc: Ec1/Ib = 250 mAdc	---	---	T(anode core)	---	175	°C
				---	---	T(anode seal)	---	150	°C
				---	---	T(base seal)	---	125	°C
---	Overload	6	Eb = 1,000 Vdc	---	---	---	---	---	---
---	Overload end point:								
1266	Total grid current	-		---	---	Ic1	---	-25	µAdc

NOTES:

1. When the tube is operated at 100 percent of maximum rated anode dissipation at an incoming air temperature of 25° C maximum, a minimum airflow of 5.6 cubic feet per minute (cfm) at sea level shall pass through the anode cooler. The static pressure drop across the anode cooler at this flow is approximately 0.26 inch of water. If the socket on Drawing 246-JAN is used, an incoming airflow of 5.6 cfm to the grid end of the socket is required. At this flow of 5.6 cfm, the static pressure drop directly across the tube and socket is approximately 0.60 inch of water. This pressure drop varies with the amount of escaping air and with the shape and construction of the air director. The airflow rating applies at bias voltages less than 100 volts and frequencies less than 500 MHz. Air cooling of the tube shall be increased with increased negative grid bias, increased incoming air temperatures, or increased frequency of operation, or a combination. In all cases of operation, a socket which provides forced-air cooling of the base shall be used and maximum seal and radiator temperature ratings shall not be exceeded. The airflow shall be applied before or simultaneously with electrode voltages, and may be removed simultaneously with them.

2. Maximum life may be obtained by adjusting the heater voltage in accordance with the application. A table of heater voltage versus frequency is presented as a guide.

Frequency (MHz)	Ef (Vac)
Up to 300	6.00
301 to 400	5.75
401 to 500	5.50

3. In all cases of electrical tests involving application of heater voltage, the socket on Drawing 246-JAN shall be used. An incoming airflow of 6.0 cfm maximum to the grid end of the socket is permitted.

4. An infinite baffle system as shown on figure 2, or equal, with an airflow of 5.6 cfm at sea level shall be used. The static pressure drop is measured across the tube and socket.

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NOTES: -Continued

5. The forced-cooling test shall be made as follows:

At an ambient temperature of 25 C. both the base and the anode shall be cooled by applying an airflow of 5.6 cfm maximum, at sea level from a single source using the infinite baffle system as shown on figure 2, or equal.

At the specified test conditions, the anode core temperature, anode seal temperature, and the base seal temperature shall not exceed the specified limits.

All temperatures shall be measured by means of thermocouples located as follows:

Anode core: This thermocouple shall be embedded in the top of the cooler, with the anode dome removed if necessary, by means of drilling a small hole, shallow enough so that the tube vacuum shall not be lost, placing the welded thermocouple junction therein, and then peening the edges of the hole so as to hold the thermocouple firmly in place.

Anode seal: This thermocouple shall be attached, using any appropriate material, to the surface of the metal immediately above metal-to-dielectric seal.

Base seal: This thermocouple shall be attached, using any appropriate material, to the surface of the metal immediately adjacent to the base dielectric material and at the immediate periphery of the dielectric material.

In all cases, good electrical continuity between the thermocouple and the metal area in close proximity must be demonstrated before the cooling test can be performed.

6. This is a destructive test. Operate the tube in an upright position. Preheat the tube at specified test conditions for 1 minute. Remove airflow for 100 seconds. Restore airflow and after 1 minute perform the total grid current test. Any tube which evidences a short or open circuit during this test will be considered a failure.

7. The maximum value of the voltage applied to the anode and grids shall not exceed 900 volts. The pulse duration measured at 5 percent of the maximum value shall be not less than 3 microseconds (μs). At 50 percent amplitude the duration shall be less than 2 μs. The applied voltage shall have a maximum repetition rate such that the duty cycle, based on the pulse length measured at 50 percent amplitude, shall not exceed 0.0002 (0.02 percent). An alternate pulsing emission test may be used with the following conditions and limits:

Conditions	Minimum	Maximum
is = 30 a, eb = ec1 = ec2 = eid	---	850 v

For life-test end points, is = 21 a.

8. This test is to be the first test performed at the conclusion of the holding period.

9. Circuit and cavity shall be in accordance with Drawing 223-JAN.

10. This test shall be made a minimum of 30 minutes after Ef is turned off. Rated airflow shall be maintained during the 30-minute interval. Measurement to be made with General Radio Megohmeter Model No. 1862C, or equivalent. Unused elements shall be left floating.

11. Tube type 7035 has been deleted from this tube specification sheet; use MIL-E-1/1331, type 7609.

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Custodians:
Army - EL
Navy - EC
Air Force - 80

Review activities:
Air Force - 17
DSA - ES

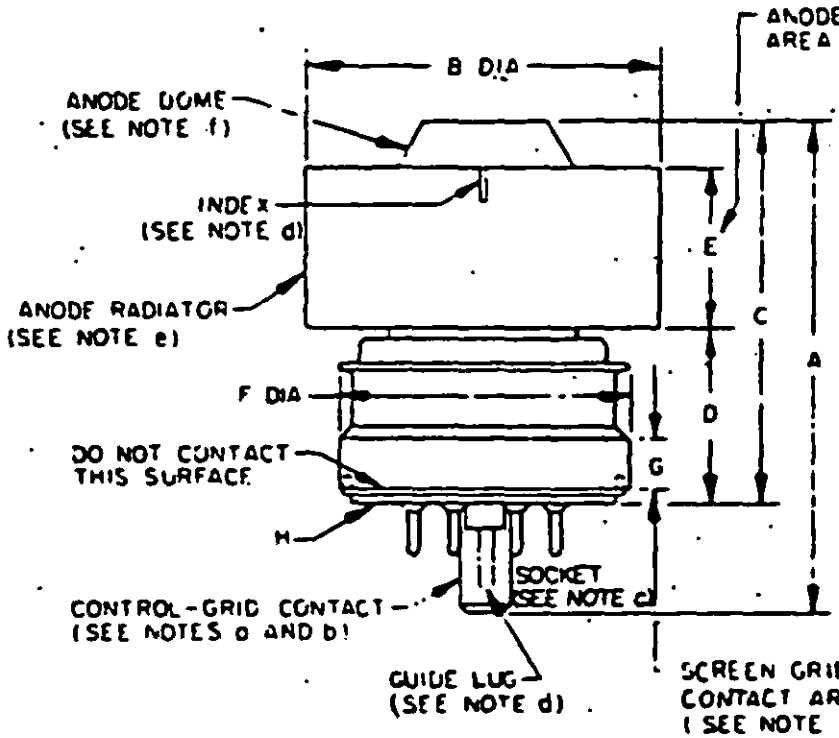
User activities:
Army - WC
Navy - AS, OS, MC, CG, SH
Air Force - 11

Preparing activity: Army - EL

Agent: DSA - ES

(Project 5960-2772)

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PIN CONNECTIONS	
Pin No.	Element
1	g2
2	k
3	k
4	k
5	int con
6	k
7	k
8	k
center post	g1
radiator	a
base ring	g2

Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 2		
A	2.224 (56.49)	2.414 (61.32)
C	1.710 (43.43)	1.860 (47.24)
Quality conformance inspection, part 3 (periodic check) (See note g)		
B	1.610 (40.89)	1.640 (41.66)
D	.750 (19.05)	.810 (20.57)
E	.710 (18.03)	.790 (20.07)
F		1.406 (35.71)
G	.187 (4.75)	
H	Base: B8-236 (EIA) (see note a)	

NOTES:

a. Pin alinement shall be checked by means of JEDEC gage GB8-3. Dimensions of control-grid contact shall be inspected by means of gages specified on Drawing 246-JAN and shall be quality conformance inspection, part 2.

b. Alinement of anode, screen-grid, and control-grid contact surfaces shall be determined by means of gage specified on Drawing 168-JAN. Quality conformance inspection, part 2, shall apply.

c. Air system socket shall be as specified on Drawing 246-JAN.

d. Location of guide lug of control-grid contact may be referenced by a notch or arrow on the anode radiator in the position shown.

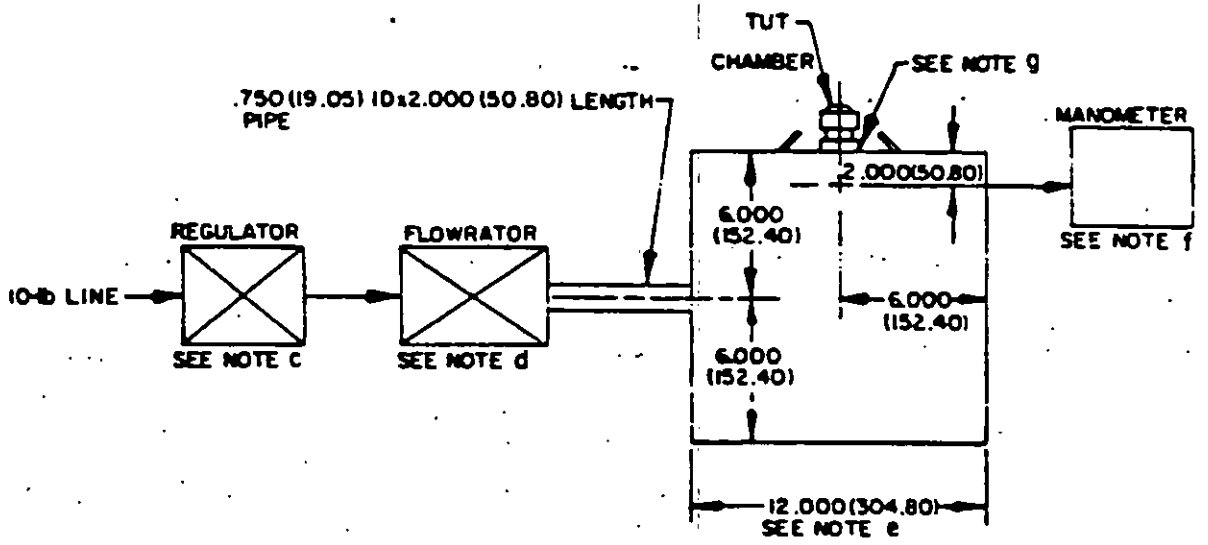
e. Anode clamping shall be confined to the anode radiator.

f. Dome contour shall be such that it will be contained within crosshatched area of template shown on figure 3. Optical comparator techniques are normally used for this purpose.

g. Perform on a sample of 10 tubes from the first production lot of each year, with one failure allowed. In case of a sample failure, the failing dimension(s) shall become part of quality conformance inspection, part 2 for three consecutive successful submissions at which time the test may revert to the yearly periodic basis.

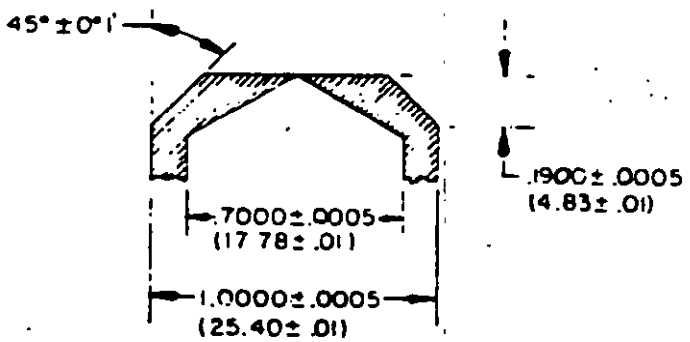
FIGURE 1. Outline drawing of electron tube type 7034.

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NOTES:
a. Dimensions are in inches.
b. Metric equivalents are in parentheses.
c. Fisher-Governer, pressure regulator model 67, or equal.
d. Fisher Porter flowrater model B4-27-10/77, or equal.
e. 12 inch (304.80 mm.) cube inside dimensions, compound sealed, or equal.
f. F.W. Dwyer manometer, 0 to 1 inch (25.40 mm) of water (Fisher Scientific Company 11-295-5 draft gage), or equal.
g. Socket specified on drawing 246-JAN.

FIGURE 2. Block diagram.



NOTES:
a. Dimensions are in inches.
b. Metric equivalents are in parentheses.

FIGURE 3. Contour limits of dome shape.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-E-1/160L ELECTRON TUBE, POWER 1/ TYPE 7034

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
   ☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 64 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.
ESC-PM 1068-68

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